

WestQuant Representation Stack (Combined)

Slide 1 — Project Overview

Problem: Quantum algorithm design typically performs one fixed translation from optimization problem to hardware. We argue representat

Approach: composable pipeline: $P \rightarrow H_i$ (Hamiltonian Explorer) $\rightarrow R_{ij}$ (Representation Scheduler) $\rightarrow Q_{ijk}$ (QoolQit compilation + emula

Results: Flagship experiment (6 problems x 5 H x 9 R x 3 replicates = 810 observations): H×R interaction dominates ($\eta^2 = 48.2\%$), embed

Importance: The H×R interaction is the dominant effect, directly motivating joint representation search. Neither H alone nor R alone determ

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Slide 2 — QoolQit Experience

Positive aspects: QoolQit provides a clean end-to-end stack: Register, Drive, QuantumProgram, compile_to(device), LocalEmulator. The em

Challenges: DMM requires AnalogDeviceWithDMM (not AnalogDevice). max_energy profile needed for auto-rescaling. Bitstring results are C

Workflow: Successive halving (cheap metrics -> emulation) is essential for search scalability. The five-class equivalence taxonomy ensures